# ISO-1 continuation (2026-10-16): certified labels, the 8th-power mass, and the residual zero classes

Research evidence report. Study: `research/iso1_weak_classes_20261007`. Markdown version with links: `residual_model_20261016/REPORT.md`. Branch `aburan28/quirky-feynman-prppv2`, PR aburan28/crypto#1954. Host: 4-core Linux container, shared; wall times are not isolated measurements.

## Summary


1. **Certified census labels through p = 107.** A new counter returns a group order only when it is the unique Hasse-interval candidate compatible with the tested points on the curve and its quadratic twist. Reruns at p = 7, 13, 37, 41, 43, 47, 53, 59 are byte-identical to the retained CSVs, so their zero labels are exact. p = 61, 67, 71, 73, 79, 83, 89, 97, 101, 103 and 107 are new.
2. **Exact 8th-power mass (conjecture).** The number $N_8(t)$ of 8th-power Legendre parameters in a class is a closed 2-adic factor times an odd-conductor Hurwitz number. It matches exhaustive censuses at p = 7, 11, 13, 17 and the registered holdouts p = 19 and p = 23 with 0 mismatches.
3. **Random thinning.** Beyond that structure, the norm-one torus samples each class at random. The parameter-free model predicts the residual zero count at the pristine holdouts p = 53, 59 and at the forward predictions p = 61, 67, 71, 73, 79, 83, 89, 97, 101, 103, 107, with binomial dispersion $1.00 \pm 0.01$.

Beyond the proved obstruction $v_2(f_\pi) \geq 2$, whether a class is weak is a draw with probability $1 - \left(1 - \frac{8}{p^2-1}\right)^{N_8 \frac{t}{12}}$. The evidence does not support any compact deterministic criterion beyond that.


## 1. Certified point counts

$Q = p^6$, so the Hasse interval is exactly $[Q + 1 - 2p^3, Q + 1 + 2p^3]$. Every curve has full rational 2-torsion, so $4 \mid \#E$, and the twist has order $2Q + 2 - \#E$. A full baby-step/giant-step scan of $4P$ lists every candidate. Further points on the curve and the twist filter the list, and the order is returned only when one candidate remains. Source: `census_runtime/src/certified.rs`. Unit test: 12 random p = 7 curves against brute-force counts.

| p | point counts | extra points | wall s | result |
|---|---|---|---|---|
| 7 | 213 | 5 | 0.01 | identical to exact square-table census |
| 13 | 2,423 | 8 | 0.2 | identical to retained CSV |
| 37 | 156,523 | 16 | 32 | identical |
| 41 | 235,901 | 20 | 56 | identical |
| 43 | 285,363 | 27 | 110 | identical |
| 47 | 407,193 | 35 | 118 | identical |
| 53 | 658,243 | 27 | 347 | identical |
| 59 | 1,010,651 | 23 | 941 | identical to retained p59 run |
| 61 | 1,154,751 | 41 | 1,265 | new: 0 / 111,630 depth-1 weak; 546 / 111,630 high-depth zero |
| 67 | 1,680,383 | 30 | 1,448 | new: 0 / 148,138 depth-1 weak; 640 / 148,136 high-depth zero |
| 71 | 2,118,901 | 34 | resumed | new: 0 / 176,436 depth-1 weak; 634 / 176,434 high-depth zero |
| 73 | 2,367,853 | 27 | resumed | new: 0 / 191,844 depth-1 weak; 622 / 191,844 high-depth zero |
| 79 | 3,247,401 | 35 | resumed | new: 0 / 243,400 depth-1 weak; 746 / 243,398 high-depth zero |
| 83 | 3,956,583 | 33 | resumed | new: 0 / 282,450 depth-1 weak; 798 / 282,448 high-depth zero |
| 89 | 5,230,501 | 53 | resumed | new: 0 / 348,524 depth-1 weak; 808 / 348,524 high-depth zero |
| 97 | 7,379,793 | 39 | resumed | new: 0 / 451,632 depth-1 weak; 934 / 451,632 high-depth zero |
| 101 | 8,674,251 | 66 | resumed | new: 0 / 510,050 depth-1 weak; 908 / 510,050 high-depth zero |
| 103 | 9,381,893 | 37 | resumed | new: 0 / 541,060 depth-1 weak; 884 / 541,058 high-depth zero |
| 107 | 10,926,163 | 55 | resumed | new: 0 / 606,798 depth-1 weak; 976 / 606,796 high-depth zero |

Receipts: `certified_20261016/` (comparison receipt, stderr receipts, CSV hashes, queue script and status).

## 2. Exhaustive λ census and the 8th-power mass

`iso1_lambda_census` enumerates every $\lambda$ in $F_Q$ or in a 2-power subgroup. For each it records the certified trace of $y^2 = x(x-1)(x-\lambda)$, whether $\lambda$ is a $2^k$-th power, and whether it lies in the norm-one torus $T$ (the weak curves).

- At p = 11 it agrees with the orbit census on all 666 rows. In each class, membership in $T$ thins $\Omega = \left(F_Q^*\right)^{2^e}$ at random: dispersion 1.012, with 31 zero classes against 32.4 expected.
- 32nd powers at p = 17 are not 2-adically local; 8th powers are, at every prime tested. This matches the fact, quoted here and not reproved, that the cover $\lambda = \mu^N$ is a congruence modular curve only for $N \mid 8$.

With $\Delta = t^2 - 4Q = f_\pi^2 D_K$, $k = v_2(f_\pi) \geq 2$, $s = \left(\frac{D_K}{2}\right)$ and $H_{\text{odd}}$ the Hurwitz sum with the 2-part of the conductor removed:

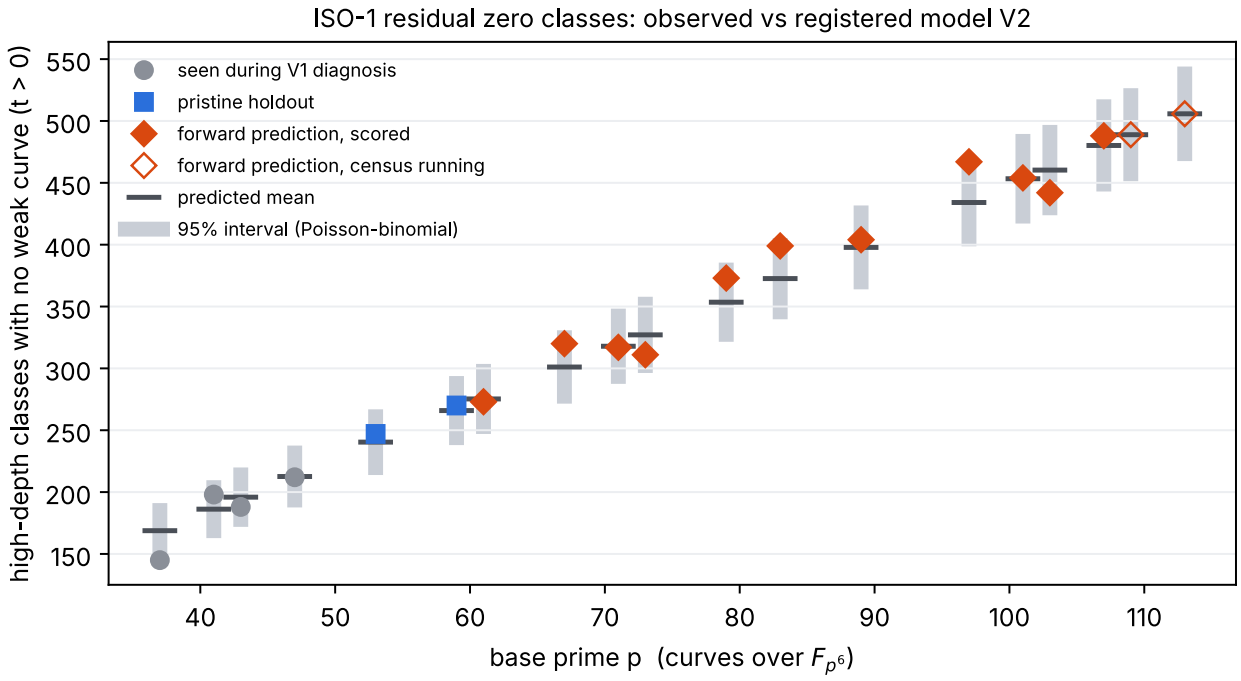
$$N_8(t) = H_{\text{odd}(t)} \cdot \begin{cases} (2-s)2^{k-2} & \text{if } t \not\equiv \pm(Q+1) \pmod{32} \\ 5(2-s)2^{k-2} - 6(1-s) & \text{if } t \equiv \pm(Q+1) \pmod{32} \end{cases}$$

| test | primes | rows | mismatches |
|---|---|---|---|
| derivation (exhaustive 8th-power census) | 7, 11, 13, 17 | 4,078 | 0 |
| registered exactness holdout | 19 | 3,249 | 0 |
| registered exactness holdout | 23 | 5,819 | 0 |

The formula is read off the tables and is not proved.

## 3. Residual zero classes: registered model V2

V1, which used class size only, was registered and failed on its training primes (dispersion about 5). V2 uses only small-prime information: weak orbits follow Binomial$\left(\frac{N_8}{12}, \frac{8}{p^2-1}\right)$, with nothing fitted. It is evaluated natively by `iso1_residual_model`.




Certified censuses (certified_20261016); predictions from class data only. Source: residual_model_20261016/model_v2_results.txt, predictions_v2.txt

Figure 1: Observed high-depth zero classes ($t > 0$) against registered V2 predictions with 95% Poisson-binomial intervals. Filled markers: certified censuses (grey: seen during V1 diagnosis; blue: pristine holdouts; orange: forward predictions, scored). Hollow: forward predictions whose census is running. Data: `figures/zero_rows.csv`; sources `model_v2_results.txt`, `predictions_v2.txt`.

| p | role | rows | zeros | predicted [95%] | disp. | AUC |
|---|---|---|---|---|---|---|
| 37 | seen | 12,321 | 145 | 168.8 [146.5, 191.1] | 0.981 | 0.959 |
| 41 | seen | 16,810 | 198 | 186.2 [162.8, 209.6] | 0.998 | 0.969 |

| p | role | rows | zeros | predicted [95%] | disp. | AUC |
|---|---|---|---|---|---|---|
| 43 | seen | 19,414 | 188 | 195.9 [171.9, 219.9] | 1.003 | 0.972 |
| 47 | seen | 25,403 | 212 | 212.6 [187.6, 237.6] | 0.997 | 0.978 |
| 53 | pristine holdout | 36,517 | 247 | 240.4 [213.8, 266.9] | 1.005 | 0.982 |
| 59 | pristine holdout | 50,474 | 270 | 265.9 [238.0, 293.8] | 1.001 | 0.986 |
| 61 | forward | 55,815 | 273 | 275.3 [247.0, 303.6] | 0.992 | 0.984 |
| 67 | forward | 74,068 | 320 | 301.1 [271.5, 330.8] | 0.996 | 0.987 |
| 71 | forward, e = 4 | 88,217 | 317 | 317.9 [287.5, 348.3] | 1.000 | 0.989 |
| 73 | forward, e = 4 | 95,922 | 311 | 327.1 [296.3, 357.9] | 1.011 | 0.989 |
| 79 | forward, e = 5 | 121,699 | 373 | 353.5 [321.5, 385.5] | 0.998 | 0.993 |
| 83 | forward, e = 3 | 141,224 | 399 | 372.6 [339.7, 405.4] | 1.001 | 0.993 |
| 89 | forward, e = 4 | 174,262 | 404 | 397.8 [363.9, 431.7] | 1.000 | 0.995 |
| 97 | forward, e = 6 | 225,816 | 467 | 434.1 [398.7, 469.6] | 1.001 | 0.996 |
| 101 | fresh, e = 3 | 255,025 | 454 | 453.3 [417.1, 489.5] | 0.995 | 0.996 |
| 103 | fresh, e = 4 | 270,529 | 442 | 460.3 [423.8, 496.8] | 1.004 | 0.996 |
| 107 | fresh, e = 3 | 303,398 | 488 | 480.2 [443.0, 517.5] | 1.000 | 0.996 |

p = 37 lies 1.5 below its interval; every other prime is inside. After p = 97 the pooled excess over the ten scored primes 53–97 was +95.3 zero rows (sd 49.8, z = 1.91). It was not localized in the tails and not isolated by e = $v_2(p^2-1)$, so it was registered as a follow-up test on fresh primes 101, 103, 107, 109 and 113 (PROTOCOL_V2.md, commit ff9e0a3f8, made before p = 101 completed): V2 is rejected as a parameter-free description if their pooled one-sided z exceeds 2.0. Three of the five are scored: p = 101 (454 against 453.3), p = 103 (442 against 460.3) and p = 107 (488 against 480.2) give a pooled excess of −9.8 with standard deviation 32.4 (z = −0.30), so no excess. Over all thirteen scored primes the pooled excess is +85.5 (sd 59.4, z = 1.44), down from 1.91 at p = 97. The predictions for the remaining fresh primes 109 and 113 (488.9 and 505.8) were committed in fa2425d14 before the p = 107 census completed. $h_K$ alone ranks zero rows with AUC 0.71–0.78.

## 4. Scope and limits

- **Proved** (`THEOREM.pdf`): no ordinary class with $v_2(f_\pi) = 1$ is weak, for every odd $p$.
- **Exact** for $p \leq 107$: the certified class labels.
- **Conjecture**, exact on six primes: the $N_8$ formula.
- **Measured**: the binomial thinning model, at two pristine holdouts and eleven forward predictions. It describes the data statistically and decides no individual class. It does not show that no exotic arithmetic criterion exists.
- Class numbers come from PARI/GP 2.15.4 `quadclassunit`, which is GRH-conditional for large $|D_K|$. This affects the predictions, not the census labels.
- Pending: the registered follow-up test (fresh primes 109 and 113; predictions committed before p = 107 completed) and the censuses for p = 109–199. p = 71–107 ran across several container sessions with checkpointing, so it has no single wall time. p = 199 alone needs about 4 days on 4 threads.